

DAY 01

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Q1. Which of the following is the largest fraction?

$\frac{6}{7}$, $\frac{5}{6}$, $\frac{7}{8}$, $\frac{4}{5}$

a. $\frac{6}{9}$ b. $\frac{4}{5}$ c. $\frac{5}{6}$ d $\frac{7}{8}$

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Q2. Which of the following fraction is the smallest?

$\frac{7}{6}$, $\frac{7}{9}$, $\frac{4}{5}$, $\frac{5}{7}$

a. $\frac{7}{6}$ b. $\frac{7}{9}$ c. $\frac{4}{5}$ d. $\frac{5}{7}$

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Q3. Which of the following is correct?

(1) $\frac{2}{3} < \frac{3}{5} < \frac{11}{15}$

(2) $\frac{3}{5} < \frac{2}{3} < \frac{11}{15}$

(3) $\frac{11}{15} < \frac{3}{5} < \frac{2}{3}$

(4) $\frac{3}{5} < \frac{11}{15} < \frac{2}{3}$

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Q4. Arrange $\frac{4}{5}$, $\frac{7}{8}$, $\frac{6}{7}$, $\frac{5}{6}$ in the ascending order:

a. $\frac{4}{5}, \frac{7}{8}, \frac{6}{7}, \frac{5}{6}$

b. $\frac{5}{6}, \frac{6}{7}, \frac{7}{8}, \frac{4}{5}$

c. $\frac{4}{5}, \frac{5}{6}, \frac{6}{7}, \frac{7}{8}$

d. $\frac{7}{8}, \frac{6}{7}, \frac{5}{6}, \frac{6}{7}$

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Q5. The Value of

$$\frac{\frac{2}{7} \times \frac{28}{11} of \frac{121}{14} - \frac{3}{7} of \frac{8}{9}}{\frac{3}{5} \times \frac{15}{8} of \frac{80}{9} - \frac{7}{3} \times \frac{15}{49} of \frac{86}{15}} \; is:$$

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Q6 TCS NQT 2021

$$1 \div \frac{3}{7} \text{ of } (6 + 8 \times \overline{3 - 2}) + [\frac{1}{5} \div \frac{7}{25} - \{\frac{3}{7} + \frac{8}{14}\}]$$

a. $\frac{5}{42}$ b. $-\frac{5}{42}$

c. 1 d. *None of these*

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Q7. The Value of

$$\left[\frac{(0.63)^2+(0.8)^2+(0.21)^2}{((0.063)^2+(0.08)^2+(0.021)^2)}\right] \div 3\frac{1}{3} \times 5\frac{1}{3} \text{ of } \frac{1}{8}, \text{ is:}$$

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Q8.

$$13\frac{3}{4} + 17\frac{2}{7} + 31\frac{1}{4} + 15\frac{5}{7} + 12\frac{2}{3} = ?$$

a. $90\frac{3}{4}$ b.$88\frac{3}{2}$ c)$90\frac{2}{3}$

d) None of these

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Q9. Simplified to:

$$99\frac{1}{7} + 99\frac{2}{7} + 99\frac{3}{7} + 99\frac{4}{7} + 99\frac{5}{7} + 99\frac{6}{7}$$

a. 102 b.104 c. 597 d. None of these

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Q10. Simplified to:

$$27\frac{3}{11} + 118\frac{2}{11} - 32\frac{5}{22} = 11\frac{6}{11} + x$$

a. $102\frac{7}{22}$ b.$102\frac{15}{22}$

c) $101\frac{15}{22}$ d) None

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Q11. What value should come in the place of question mark in following question

$$\sqrt{? + 27 * 4 + 119} = 14\frac{2}{3} + 6\frac{1}{3}$$

a. 176 b. 214 c. 252 d. 188 e. 212

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Q12. $\frac{1}{8.33\%}$ of $\frac{810}{4} \div 360 \times \frac{40}{9} = ?$

a. 10
b. 20
c. 30
d. 60
e. NOT

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Q13. $\sqrt{81\%}$ of 800 + $\sqrt[3]{51.2\%}$ of 600 = ?

a. 120 b. 600 c. 720 d. 1200 e. NOT

Q14

$$[(16)^{-1/200\%} - (30\%)^{200\%}]^{50\%} = ?$$

a. 0.04
b. 0.2
c. 0.4
d. 0.16
e NOT

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Q15. The value of

$$1+\cfrac{1}{1+\cfrac{1}{1+\cfrac{1}{1+\cfrac{1}{1+\cfrac{2}{3}}}}}$$

a. $\frac{21}{13}$ b. $\frac{17}{2}$ c. $\frac{34}{21}$ d. $\frac{8}{5}$

Q16 find the value of $2x + \frac{7}{4}$

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If $x = 1 + \cfrac{1}{1 + \cfrac{1}{1 + \cfrac{1}{1 + \cfrac{1}{2}}}}$

$2 \times \frac{13}{8} + \frac{7}{4} = \frac{20}{4} = 5$

a. 5

b. $\frac{13}{8}$

c. 4

d. $\frac{7}{8}$

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Q17 . Find the value of

$$1-\cfrac{1}{1-\cfrac{1}{1-\cfrac{1}{1-\frac{2}{3}}}} =$$

-2 -1

2 3 1 -2 -3 -1

-3 -(-2)

≠1

≠3

a. $\frac{2}{3}$ b. $-\frac{1}{3}$ c. $\frac{1}{3}$ d. $-\frac{2}{3}$

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Q18. The value of

$$\left(1+\frac{1}{2}\right)\left(1+\frac{1}{3}\right)\left(1+\frac{1}{4}\right)\ldots\ldots\ldots\ldots\left(1+\frac{1}{120}\right)$$

a. 30 b. 40.5 c. 60.5 d.121

$$\frac{3}{2}\times\frac{4}{3}\times\frac{5}{4}\cdots\cdots\cdots\frac{119}{118}\times\frac{120}{119}\times\frac{121}{120}$$

$$\frac{121}{2}=60.5$$

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Q19. The simplified value of

$$\left(1-\frac{1}{3}\right)\left(1-\frac{1}{4}\right)\left(1-\frac{1}{5}\right)\ldots\left(1-\frac{1}{99}\right)\left(1-\frac{1}{100}\right)$$

$a.\ \frac{2}{99}$ b. $\frac{1}{25}$ c. $\frac{1}{50}$ d. $\frac{1}{100}$

Hint.

Q20. $\left(1+\frac{1}{2}\right)\left(1+\frac{1}{4}\right)\left(1+\frac{1}{6}\right)\left(1+\frac{1}{8}\right)\left(1-\frac{1}{3}\right)\left(1-\frac{1}{5}\right)\left(1-\frac{1}{7}\right)$ $= \left(1+\frac{1}{x}\right)$ then what is the value of x?

a. 6 b. 8 c. 5 d. 7

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Q21. Consider a sequence, where the nth term is

$t_n = \frac{n}{(n+2)}$, n=1, 2..

The value of $t_3 * t_4 * t_5 * \ldots t_{53}$ equals

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a. $\frac{2}{495}$ b. $\frac{2}{477}$ c. $\frac{12}{55}$

d. $\frac{1}{1485}$ e. $\frac{1}{2970}$

H.W.

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Q22. The sum of the first 20 terms of the series

$$\frac{1}{5\times6}+\frac{1}{6\times7}+\frac{1}{7\times8}+\ldots\ldots+\frac{1}{24\times25}$$ is:

$\frac{1}{5}-\frac{1}{6}+\frac{1}{6}-\frac{1}{7}+\frac{1}{7}-\frac{1}{8}+\cdots\cdots-\frac{1}{24}+\frac{1}{25}$

$\frac{1}{5}-\frac{1}{25}=\frac{4\times4}{25\times4}$

$\frac{1}{5\times4}=\frac{1}{5}-\frac{1}{6}$

b $\frac{16}{100}$

a. 1.6
b. 0.16 ✓
c. 0.016
d. 0.0016

$1 = 1$

$1+1 = 1\times 2$

$1+1+1 = 1\times 3$

$1+1+1+\dots = 1\times n$

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Q23. Find sum of

$$\left(1-\frac{1}{n+1}\right)+\left(1-\frac{2}{n+1}\right)+\left(1-\frac{3}{n+1}\right)+\dots\dots+\left(1-\frac{n}{n+1}\right)$$

a. n b. $\frac{1}{2}n$ c. (n+1) d. $\frac{1}{2}(n+1)$

$$[1+1+1+\dots n]-\frac{1}{n+1}[1+2+3+\dots n]$$

$$n-\frac{1}{n+1}\left[\frac{1}{2}n(n+1)\right]$$

$$n-\frac{n}{2}=\frac{n}{2}$$

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"Question of the day"

Q24. One page is torn from a booklet whose pages are numbered in the usual manner starting from the first page as 1. The sum of the numbers on the remaining pages is 195. The torn page contains which of the following numbers?

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a) 5, 6

b) 7, 8

c) 9, 10

d) 11, 12

n = 20 pages

1 2 3 4 5 6 7 8 9 10

Sum $= \frac{1}{2} n(n+1) = \frac{1}{2} \times 20 \times 21$

$= 210$

$210 - 195$

$= 15$

xx

elitmus

$\frac{1}{2} \times 19 \times 20^{10}$

190

[CSAT]

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MBA

Plant

$1+2+3 = \frac{1}{2}n(n+1)$

$= \frac{1}{2} \times 3 \times 4 = 6$

Q25.

Find the Average of :

1 , 2 , 2 , 3, 3 , 3 , 4, 4, 4, 4, ...10,10,10,10,10(10 times)

6

1 , 2 , 2 , 3 , 3 , 3 , 4 , 4 , 4 , 4 , 5 , 5 ——

$\frac{1}{6}n(n+1)(2n+1)$

$1+4+9+16+25+\cdots-100$

$Avg = \frac{Sum}{No} =$

$\frac{1}{2}n(n+1)$ ← $1+2+3+4+5+\cdots-10$

Comments:

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Q26.

Which value among 3^{200}, 2^{300} and 7^{100} is the largest?

a. 3^{200} **b. 2^{300}** **c. 7^{100}** **d. All area equal**

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Q27. The greatest number among 2^{60}, 3^{48}, 4^{36}, and 5^{24} is:

a. 2^{60} b. 3^{48} c. 4^{36}, d. 5^{24}

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Q28. Which one among $\sqrt[3]{6}$, $\sqrt[2]{5}$ and $\sqrt[6]{12}$ is the largest?

a. $\sqrt[3]{6}$ **b. $\sqrt[2]{5}$** **c. $\sqrt[6]{12}$** **d. All are equal**

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Q29. Which value among $\sqrt[4]{7}$, $\sqrt[3]{11}$ and $\sqrt[12]{1257}$ is the largest?

a. $\sqrt[3]{11}$ b. $\sqrt[4]{7}$ c. $\sqrt[12]{1257}$ d. All are equals

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Q30. The greatest among the numbers $\sqrt[2]{8}, \sqrt[4]{13}, \sqrt[5]{16}, \sqrt[10]{41}$ is:

a. $\sqrt[4]{13}$ b. $\sqrt[5]{16}$ c. $\sqrt[10]{41}$ d. $\sqrt[2]{8}$

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Q31. $\sqrt{121} + \sqrt{12321} + \ldots\ldots\ldots + \sqrt{1234567654321} =$

a. 1234567 b. 1234566 c. 1234565 d. 12345677

Q32. $\sqrt{3\sqrt{3\sqrt{3}\ldots.}}$ is equal to.

a. $\sqrt{3}$ b. 3 c. $2\sqrt{3}$ d. $3\sqrt{3}$

Q.33

$$\left[(68921)^{1/3} - (2744)^{1/3}\right]^{1/3} = ?$$

a. 2
b. 3
c. 4
d. 5
e. *None of these*

Q26. Find the value of $\sqrt{3+\sqrt{3+\sqrt{3+\sqrt{3+\ldots\ldots\infty}}}}$

a. 3 b. 4 c. 6 d. 5